

Literature Review

INTELLIGENT SWARM BASED BOTS FOR CONTINUOUS AREA PATROLLING AND DETECTION

1. Malathy et al. - Energy-Efficient Clustered Wireless Sensor Networks for Border Surveillance

Malathy and colleagues investigate energy-efficient clustered wireless sensor network (WSN) architectures for border surveillance, integrating ant colony optimization (ACO) algorithms for optimal routing performance. Their work addresses the fundamental challenge of balancing computational efficiency, energy consumption, and data security in resource-constrained environments requiring continuous operation. By organizing sensor nodes into hierarchical clusters, the network significantly reduces redundant transmissions and distributes energy consumption evenly, preventing premature node failure. The ACO-based routing protocols mimic ant colony foraging behavior to dynamically discover and maintain optimal data transmission paths. A critical contribution is the integration of elliptic curve cryptography (ECC) to secure transmissions without excessive computational overhead on resource-limited nodes. Unlike traditional cryptographic methods such as RSA, ECC provides equivalent security with significantly smaller key sizes, making it well-suited for WSN applications. Simulation results validate extended network lifetime, high packet delivery ratios, and suitability for long-term border monitoring.

2. Essendorfer et al. - Integrated Multi-Sensor Data Fusion for Border Control

Essendorfer and colleagues present an integrated framework for border surveillance emphasizing heterogeneous sensor data fusion and transnational information sharing to enhance situational awareness. Their work addresses the complexity of modern border security where infrared cameras, electro-optical systems, radar, and sonar must operate cohesively across vast terrain. The core innovation lies in a unified data fusion architecture integrating disparate sensor streams into a coherent operational picture. The SOBCAH project implements this vision through a shared database enabling real-time information exchange among border agencies across multiple countries. Image processing modules extract features from raw sensor data, fused using probabilistic reasoning to generate actionable intelligence. This multi-sensor approach significantly reduces false alarms while improving detection probability, as anomalies must be corroborated across multiple independent sensing modalities before triggering alerts. The system provides intuitive visual displays overlaying fused sensor information on geographic maps, demonstrating effective interoperability critical for comprehensive border control.

3. Sauter et al. - Swarming Control for Unmanned Vehicles in Surveillance

Sauter and colleagues demonstrate swarming control algorithms for unmanned ground and aerial vehicles applied to surveillance and infrastructure protection. Their digital pheromone-based approach enables autonomous, adaptive coordination of heterogeneous sensor

platforms, supporting complex behaviors including target tracking and multi-sensor fusion. The system mimics biological swarm intelligence where individual agents follow simple rules to achieve sophisticated collective behavior without centralized control. Real-world tests show how swarms of UAVs, UGVs, and sensor networks interact to detect and track intruders while cooperating with human patrols. The digital pheromone mechanism allows agents to communicate indirectly through environmental markers, creating dynamic spatial patterns that guide swarm movement and task allocation. The paper demonstrates robustness against individual agent failures, scalability to large deployments, and effective operator interface design for commanding large swarms in security scenarios, highlighting practical implementation considerations for autonomous surveillance systems.

4. Fatima et al. - IoT-Based Smart Border Security with Machine Learning

Fatima and colleagues propose an IoT-based smart border security system utilizing machine learning for intruder detection. Employing sensors, thermal imaging, and camera data analyzed by Faster R-CNN and EfficientDet models, the system distinguishes between military personnel and intruders with high accuracy. The architecture integrates edge computing for real-time processing with cloud-encrypted storage for long-term data retention and analysis. Automatic SMS alerts notify security personnel immediately upon threat detection, enabling rapid response. The system mitigates false alarms through multi-stage verification and improves resource efficiency by focusing attention on genuine threats. The paper discusses practical deployment challenges in rugged terrains with limited manpower, addressing issues of power supply, network connectivity, and environmental robustness. Cloud integration enables scalable remote monitoring and centralized command-and-control operations, allowing security forces to manage extensive border regions from consolidated facilities while maintaining situational awareness across distributed sensor deployments.

5. Arjun et al. - Wireless Sensor Networks for Border Surveillance

Arjun and colleagues survey multiple wireless sensor network (WSN) techniques for border surveillance and intruder detection, highlighting hybrid systems integrating seismic, acoustic, infrared, and camera sensors to detect human intrusions in diverse border scenarios. The paper discusses multi-layered architectures and communication protocols critical for large-scale deployments, examining how different sensor modalities complement each other to provide comprehensive coverage. Seismic sensors detect ground vibrations from footsteps or vehicles, acoustic sensors capture sound signatures, infrared sensors identify heat patterns, and cameras provide visual confirmation. The authors analyze challenges including false alarms from wildlife or environmental factors, sensor coverage limitations, and energy constraints. They propose multi-sensor fusion approaches to improve detection accuracy by correlating data across modalities, reducing false positives while maintaining high detection rates. The survey emphasizes the importance of intelligent data aggregation and energy-efficient communication protocols to extend network lifetime and provide early warnings, enhancing overall border security effectiveness through coordinated sensor operation.

6. Thomas et al. - Machine Learning Approaches Survey

Thomas and colleagues present a comprehensive survey on machine learning approaches, categorizing them into supervised, unsupervised, semi-supervised learning, and reinforcement learning methods. The paper explains key algorithms including decision trees, rule-based classifiers, naive Bayes, k-nearest neighbors, neural networks, and support vector machines, providing foundational understanding of their operational principles. For supervised learning, they discuss how labeled training data enables algorithms to learn input-output mappings for classification and regression tasks. Unsupervised methods cluster unlabeled data to discover hidden patterns and structures. Semi-supervised approaches leverage small amounts of labeled data with larger unlabeled datasets, addressing practical scenarios where labeling is expensive. Reinforcement learning enables agents to learn optimal behaviors through trial-and-error interaction with environments. The paper discusses applications across domains like pattern recognition, natural language processing, computer vision, and autonomous systems, offering insights directly relevant for AI-driven border surveillance and detection systems that require robust, adaptive decision-making capabilities.

7. Nahata et al. - YOLO-Based Real-Time Border Security

Nahata and colleagues present a YOLO-based real-time border security surveillance system integrating deep learning with camera and UAV inputs to detect humans, vehicles, drones, and weapons. The paper highlights challenges in traditional border monitoring including limited visibility, vast coverage areas, and manual patrol inefficiencies. Their automated object detection approach enables timely threat recognition with improved detection speed and accuracy compared to manual observation. The YOLO (You Only Look Once) architecture performs single-stage detection, processing entire images in one forward pass for real-time performance crucial in dynamic border environments. The system reduces reliance on manpower and enables scalable surveillance across extended perimeters. An integrated alert mechanism provides rapid notification to security personnel when threats are detected, including classification of threat types to prioritize responses. The architecture supports integration with IoT devices for extended coverage, enabling coordinated multi-sensor deployments. The study demonstrates practical feasibility for deployment in resource-constrained border regions requiring continuous automated monitoring.

8. Mahjoub et al. - Autonomous Robot Swarm for Border Intrusion

Mahjoub and colleagues develop an autonomous wireless sensor network employing a swarm of mobile robots for border intrusion detection. Mimicking natural swarm behavior, their heterogeneous robot system collaboratively navigates complex terrains using distributed coordination algorithms that enable emergent collective intelligence without centralized control. The robots employ Yolov5 for object identification, providing real-time visual recognition of humans, vehicles, and other entities of interest. ROS (Robot Operating System) and Gazebo simulation environment are utilized for system design, testing, and coordination protocol development before physical deployment. The study emphasizes cooperative navigation strategies where robots share information about obstacles, terrain

features, and detected targets to optimize coverage and tracking efficiency. Robust multi-robot communication protocols ensure system resilience against individual node failures and communication disruptions. Fault tolerance mechanisms enable continued operation even when subset of robots malfunction, maintaining surveillance coverage through dynamic task reallocation. The research showcases potential for scalable, resilient continuous border surveillance through autonomous mobile platforms.

9. Reddy et al. - Mini Seeker Robot for Border Surveillance

Reddy and colleagues propose the Mini Seeker Robot, an automated border surveillance platform equipped with ultrasonic sensors, facial recognition, and live video streaming functionalities. Controlled via Raspberry Pi microcontroller and integrated with OpenCV computer vision library, the robot detects obstacles autonomously and identifies individuals by matching facial data with pre-registered databases. The paper addresses challenges in persistent border monitoring where human patrols face fatigue, limited coverage, and exposure to dangerous conditions. The cost-effective solution operates autonomously following pre-programmed patrol routes or under remote manual control when operator intervention is required. Ultrasonic sensors enable collision avoidance and terrain mapping for safe navigation. Facial recognition provides identity verification capabilities, distinguishing authorized personnel from potential intruders. Live video streaming allows remote operators to assess situations in real-time and make informed decisions. The platform enhances security while reducing human risk exposure in border regions, demonstrating practical applicability for supplementing traditional patrol methods with robotic assistance.

10. Sathe et al. - Helmet Detection and Number Plate Recognition

Sathe and colleagues focus on helmet detection and number plate recognition using deep learning models YOLOv5 and EasyOCR within traffic surveillance contexts. Their methodology involves bounding box overlap techniques to associate detected helmets with corresponding motorcyclists, effectively identifying helmet violation instances for traffic law enforcement. The study achieves high accuracy through transfer learning, leveraging pre-trained models and fine-tuning on domain-specific datasets. Comprehensive dataset augmentation including rotation, scaling, brightness adjustment, and synthetic variations improves model robustness across diverse lighting and weather conditions. EasyOCR performs optical character recognition on license plates, enabling vehicle identification and tracking. The techniques demonstrate relevance to real-time object detection and multi-class classification applicable to security applications including border monitoring where vehicle identification and occupant analysis are critical. The integration of detection and recognition provides comprehensive situational awareness, enabling automated enforcement and alerting systems that reduce manual monitoring burden while maintaining high accuracy across varied operational conditions.

11. Noor et al. - Digital Border Surveillance System using YOLOv7

Noor and colleagues propose a Digital Border Surveillance System (DBSS) using YOLOv7 for real-time object detection to monitor illegal migration and trafficking across borders. The system integrates IoT sensors including PIR (Passive Infrared) and ultrasonic sensors with a custom dataset of 7347 images featuring uniformed personnel, civilians, and animals for comprehensive detection capability. The model achieves 89% alarm success rate and 85% notification accuracy, demonstrating reliable operational performance. The system implements intelligent threat-level differentiation: severe alarms for detected civilians potentially indicating illegal crossings, moderate alerts for unknown uniforms suggesting unidentified personnel, and minor notifications for animals to avoid false positives. Lateral movement filtering ignores non-threatening parallel motion along the border, focusing attention on perpendicular crossing attempts. A centralized control room dashboard provides video logging, real-time monitoring, and notification management, enabling coordinated response across multiple surveillance zones. The architecture supports scalable deployment across extensive border regions with distributed sensor networks feeding into unified command infrastructure.

12. Kumar et al. - Thermal Image Pedestrian Detection

Kumar and colleagues develop a deep learning framework using YOLOv5 variants for pedestrian detection in thermal images, addressing limitations of RGB-based systems in challenging weather conditions including fog, rain, and nighttime operations. Trained on the UNIRI-TID thermal imaging dataset containing 7412 annotated images, YOLOv5-S achieves the highest mean Average Precision (mAP) of 0.9908, outperforming tYOLOv3 across various thermal benchmarks. The model requires less GPU and CPU resources, making it suitable for IoT deployment on resource-constrained edge devices in remote border locations. The study examines effects of layer freezing and data augmentation strategies, showing that backbone freezing accelerates training with minimal accuracy loss by retaining learned low-level features while adapting high-level detection layers. This enables efficient model updates and domain adaptation. The research demonstrates particular relevance for border surveillance in deserts and mountainous regions where visual spectrum cameras perform poorly due to low light, dust, or adverse weather, while thermal imaging reliably detects human heat signatures regardless of lighting conditions.

13. Taneja et al. - YOLO-Based Surveillance with Automated Alerts

Taneja and colleagues present a YOLO-based real-time surveillance system detecting persons, vehicles, weapons, and drones, replacing labor-intensive patrols with automated threat identification and email alerts via Flask web framework integration. The methodology employs dataset annotation using Roboflow platform, enabling efficient labeling and preprocessing of training data. YOLO training provides single-stage detection with superior speed over two-stage methods like Faster R-CNN, critical for UAV and border applications requiring real-time processing. The system implements OAuth2-secured email notifications, ensuring authenticated and encrypted alert delivery to authorized personnel. Comparative

analysis highlights YOLOv8's 94.9% drone detection accuracy and 70% overall mAP across multiple object classes, demonstrating robust multi-class performance. The paper discusses future expansion including IoT deployment for distributed sensor networks and additional object classes covering broader threat categories. The automated alert system reduces response times by immediately notifying security teams upon threat detection, enabling rapid intervention while maintaining continuous monitoring without human fatigue limitations.

14. Daou et al. - IoT-Based Solar Charger for Field Deployments

Daou and colleagues utilize Arduino UNO R4 WiFi and WebSocket communication to a Flutter mobile app for real-time photovoltaic and battery parameter visualization, creating an IoT-based MPPT (Maximum Power Point Tracking) solar charger. They achieve over 97% tracking efficiency with closely matching laboratory and app readings by combining perturb- and-observe MPPT algorithm, high-precision sensing, and low-latency WebSocket telemetry. This demonstrates how tightly coupled embedded control, sensing, and real-time IoT communication can be integrated into single low-cost platforms. The work has direct application to field-deployed, power-aware surveillance assets in remote border regions where grid power is unavailable. Solar charging extends operational lifetime of sensor nodes, cameras, and communication equipment without requiring frequent battery replacement or refueling. The real-time monitoring enables predictive maintenance and energy management optimization, ensuring continuous surveillance coverage. The low-cost implementation makes large-scale deployment economically feasible across extensive border perimeters requiring distributed autonomous power systems.

15. Saponara et al. - 60 GHz CMOS-SOI Transceiver for IoT

Saponara and colleagues design essential components of a fully integrated 60 GHz CMOS-SOI transceiver for IoT nodes requiring multi-Gbps links over approximately 10 meters, targeting applications including vehicle-borne sensing systems, smart mobility, and surveillance. To meet ECMA-387-class link specifications at low power and compact area, they co-optimize wideband low-noise amplifiers (LNAs), power amplifiers (PAs), and on-chip antennas, exploiting the 57-66 GHz unlicensed band and CMOS-SOI technology advantages. The study demonstrates that compact high-data-rate nodes appropriate for dense camera and LiDAR networks are enabled by integrating RF, baseband, and antenna on single die, though continuous power consumption must be carefully managed in battery-powered deployments. The millimeter-wave frequency enables high bandwidth for streaming multiple video feeds and sensor data simultaneously from surveillance platforms. Short-range high-speed links are ideal for vehicle-to-infrastructure communication in border patrol scenarios where mobile units must rapidly transmit large data volumes to command centers or relay stations.

16. Geetha M et al. - IoT-Powered Surveillance Robots

Geetha M and colleagues present mobile robot platforms for surveillance integrating IoT connectivity with onboard sensing for remote monitoring applications. The IoT-Powered Smart Surveillance Sentinel Bot emphasizes basic autonomy in relatively straightforward environments with standard communication models, providing foundational capabilities for patrol and inspection tasks. The Model Spy Bot introduces Li-Fi (Light Fidelity) technology for low-RF-emission, light-based communication enabling short-range covert surveillance. This trades transmission range and robustness for reduced detectability and electromagnetic interference (EMI) immunity, valuable in scenarios requiring electronic silence or operation in RF-congested environments. Li-Fi's directional nature provides inherent security against interception compared to omnidirectional radio communications. Together, these studies demonstrate how low-cost robotics and alternative physical-layer links can enhance surveillance coverage in specific operational contexts. However, they only partially address multi-robot coordination, adversarial conditions, or long-range border scenarios requiring more sophisticated communication and coordination mechanisms for comprehensive security coverage.

17. Sergio Saponara et al. - Navigation Interface for Multiple Robots

Sergio Saponara and colleagues develop a user-centric navigation interface and control framework enabling grouping and coordinated motion of multiple robots in shared environments. By providing interaction modalities and algorithms for grouping, tasking, and conflict management across multiple platforms, it reduces operator workload in multi-robot missions where single operators must manage numerous autonomous agents simultaneously. The interface allows operators to define robot groups, assign collective objectives, and monitor mission progress through intuitive visualization. Conflict management algorithms prevent collisions and deadlocks when multiple robots operate in proximity or compete for shared resources. Although primarily demonstrated in indoor or structured environments, the grouping concepts, shared navigation interfaces, and conflict management directly apply to border-surveillance UGV swarms and patrol coordination. The scalable human-robot interface enables supervision of large autonomous fleets from centralized control facilities, critical for managing extensive border security operations where dozens or hundreds of robots may operate across wide geographic areas requiring coordinated patrol patterns and response protocols.

18. Alajrami et al. - AI-Based Human Detection for Storage Optimization

Alajrami and colleagues design an AI-based desktop program using OpenCV and Haar-cascade classifiers under Linux to trigger recording only when human body or face is detected, optimizing storage utilization. They report approximately 99% body-detection and 96% face-detection accuracy on test sequences by switching from motion-only triggering to human-specific detection. This reduces video storage by 27% compared to continuous recording while maintaining complete coverage of human activity. The system greatly reduces search effort and storage costs by eliminating hours of empty footage from archived

recordings. Although the system relies on comparatively traditional Haar/HOG-style detection techniques, which may be less resilient than contemporary deep learning detectors under occlusion, lighting variation, and long-range conditions, it provides practical storage optimization for large-scale surveillance deployments. The approach is particularly valuable for border surveillance where continuous recording across numerous cameras would generate prohibitive storage requirements, while event-triggered recording maintains evidential value and operational awareness with manageable data volumes and simplified forensic review.

19. Eman Alajrami et al. - Deployment Considerations for AI Surveillance

Eman Alajrami and colleagues discuss deployment factors for AI-based human identification including user acceptance, frame rate and resolution configuration, and integration with existing CCTV infrastructure. Evaluations with real organizations demonstrate that people find AI-assisted selective recording both useful and easy to use, suggesting successful integration into existing systems without requiring complete camera replacement. This is important for upgrading large surveillance networks already deployed at borders or critical facilities, where infrastructure replacement would be prohibitively expensive. The study examines practical considerations including computational requirements, network bandwidth, storage architecture, and operator training needs. User acceptance testing reveals that security personnel readily adopt AI-assisted systems when they demonstrably reduce workload without compromising security coverage. The research emphasizes backward compatibility and incremental upgrade paths, enabling organizations to modernize surveillance capabilities progressively rather than requiring disruptive wholesale replacements. These deployment insights inform practical implementation strategies for enhancing border surveillance infrastructure cost-effectively.

20. Geetha et al. - Blockchain-Based Defense Monitoring

Geetha and colleagues propose a next-generation defense monitoring system incorporating digital twin concepts, blockchain, IoT, and AI-based facial recognition for secure asset tracking and access control. The system provides immutable logs, automated responses via smart contracts, and high resilience to tampering by utilizing streaming face detection/recognition, GPS-based activity logging, and permissioned blockchain (Hyperledger Fabric). Experimental results in military-style facilities demonstrate strong resistance to simulated cyberattacks, approximately 98% face-recognition accuracy under favorable conditions, GPS localization within about 5 meters, and blockchain logging latency of 2-3 seconds. The blockchain architecture eliminates single points of failure present in centralized database systems, with distributed ledger ensuring data integrity even if individual nodes are compromised. Smart contracts automate access control decisions and alert generation based on pre-defined security rules, enabling rapid autonomous responses to security violations. The digital twin framework maintains virtual representations of physical assets and facilities, supporting predictive analytics and scenario simulation for security planning and threat assessment.

21. Mohith S et al. - Blockchain for Defense Applications

Mohith S and colleagues examine how blockchain enhances auditability and decentralization in defense monitoring systems, removing single points of failure inherent in traditional password and ID systems stored in centralized databases. The distributed ledger provides tamper-evident logs of all access attempts, movements, and security events, creating comprehensive audit trails for forensic analysis and compliance verification. Unlike conventional systems where database compromise exposes all credentials, blockchain's cryptographic linking and distributed storage make retroactive modification computationally infeasible. However, the paper highlights ongoing challenges with blockchain scalability and latency for very large deployments, particularly in real-time defense operations requiring immediate decision-making. High transaction throughput and low confirmation times are critical for operational security systems that cannot tolerate delays in access control or alert generation. The study emphasizes need for lightweight consensus mechanisms and careful architectural partitioning to balance security benefits against performance requirements, suggesting hybrid approaches combining blockchain's immutability for audit logs with conventional databases for real-time operational data.

22. Ramyashree P M et al. - Privacy-Preserving Border Surveillance

Ramyashree P M and colleagues note that conventional border-surveillance architectures frequently emphasize detection performance through cameras, ground sensors, and radar while placing little emphasis on formal privacy guarantees and secure data handling. The paper examines cryptographic primitives including encryption, secure aggregation, and privacy-preserving analytics used in surveillance and IoT streams, alongside sophisticated deep learning-based human and object detection techniques. Existing works demonstrate either strong detection accuracy or strong privacy separately, but rarely both simultaneously, resulting in either highly secure frameworks with limited real-time analytics capability or powerful but privacy-invasive systems. The paper proposes a border surveillance architecture combining deep learning-based target detection with cryptographic methods to safeguard transmitted and stored data, preserving both operational effectiveness and strict confidentiality. Secure aggregation allows multiple sensor nodes to collectively compute detection statistics without revealing individual sensor data. Homomorphic encryption enables computation on encrypted data, allowing analytics without decryption. The architecture addresses privacy concerns critical when surveillance systems must comply with data protection regulations while maintaining security effectiveness.

23. Dilbar Roshan et al. - AI-Powered Search and Rescue Rover

Dilbar Roshan and colleagues present a human detection rover for AI-powered search and rescue in disaster zones, integrating PIR motion detection with YOLO V3 object recognition and GPS tracking for locating survivors. Their system uses Raspberry Pi microcontroller with Wi-Fi/MQTT communication enabling wireless remote control and cloud-based monitoring. The system achieves 96% detection accuracy across various environments, functioning

effectively even in cluttered or low-visibility conditions where manual search is slow and unsafe. PIR sensors provide initial motion alerts triggering camera-based YOLO detection for human verification, reducing false positives from animal movement or environmental factors. GPS coordinates of detected survivors are transmitted immediately to rescue coordination centers, enabling rapid response deployment. The rover's mobility allows access to collapsed structures and hazardous areas where human entry would be dangerous. Future enhancements include thermal imaging for improved nocturnal detection and SLAM (Simultaneous Localization and Mapping) integration for autonomous navigation in complex environments without GPS availability, critical for indoor and underground search operations.

24. Greeshma Lingam et al. - Hybrid Bot Detection Framework

Greeshma Lingam and colleagues propose a hybrid bot-detection framework combining reinforcement learning and swarm-optimization concepts to improve identification accuracy in large social networks. Rather than depending solely on static rules, the model continuously learns behavioral cues from user interactions and reshapes decision boundaries over time through adaptive learning mechanisms. The authors highlight the system's adaptive nature helping identify rapidly evolving bot patterns that evade traditional detection methods. Reinforcement learning enables the system to improve detection strategies based on feedback from confirmed bot identifications and false positive corrections. Swarm optimization distributes detection across multiple agents that share information about suspicious patterns, enabling collective intelligence emergence. Findings show improved precision and reduced false positives compared to rule-based or single-model approaches, making the framework suitable for real-world deployments where bot behavior changes dynamically. While focused on social network security, the adaptive detection principles apply to surveillance systems identifying anomalous behaviors in border monitoring contexts where threat patterns evolve continuously.

25. Sudhanshu Janwadkar et al. - Portable GPS Tracking Solution

Sudhanshu Janwadkar and colleagues present a portable GPS tracking solution designed for real-time monitoring of individuals in outdoor locations. The system retrieves accurate positional coordinates and transmits them wirelessly to remote base stations enabling continuous oversight for applications including child safety, elderly assistance, and asset tracking. Through compact hardware design combining GPS receiver, microcontroller, and wireless transmitter, the system provides straightforward, cost-effective field-level tracking capability. Basic mapping functions visualize tracked positions on digital maps, enabling operators to monitor multiple targets simultaneously. The authors demonstrate reliable performance under varied outdoor conditions including urban, rural, and partially forested environments where GPS signal quality fluctuates. Power optimization techniques extend battery life for multi-day autonomous operation. The tracking system supports geofencing capabilities, generating alerts when tracked entities enter or exit predefined boundary zones, valuable for perimeter security and movement restriction enforcement in border patrol applications where detecting unauthorized crossings or monitoring authorized personnel movements is operationally critical.

26. K. Sudhakar et al. - FPGA-Based Rescue Bot

K. Sudhakar and colleagues develop an FPGA-based rescue bot for surveillance combining internet-controlled mobility with PIR sensor-based human detection and camera-based video transmission for disaster response and border monitoring applications. The robot integrates distance sensors for obstacle detection and environmental monitoring units for hazard assessment. By using FPGA's parallel processing capabilities, the system handles multiple sensor streams simultaneously without delay, crucial in unpredictable rescue conditions requiring immediate response to rapidly changing situations. FPGA implementations provide deterministic timing and hardware-level concurrency unavailable in sequential microcontroller architectures, enabling guaranteed real-time performance. The authors highlight such architectures enhancing onsite safety by allowing robots to enter areas humans cannot safely access, including structurally unstable buildings, chemical spill zones, or active conflict areas. Internet control enables remote operators to guide robots from safe distances while receiving real-time video and sensor feedback. The FPGA's reconfigurability supports field updates and mission-specific customization, adapting robot behavior to diverse operational requirements without hardware modifications.

27. Anisom Chakraborty et al. - Laser Beam Intrusion Detection

Anisom Chakraborty and colleagues propose a laser beam intrusion detection mechanism for securing outdoor perimeters across long distances. Any interruption to the laser line triggers instant alerts through an IoT platform, allowing remote monitoring personnel to assess situations quickly. With high precision and low power consumption, the system is suitable for continuous deployment along fence lines and boundary perimeters. The authors note the solution reduces false alarms compared to complex radar or camera-based perimeter systems, as laser beam interruption provides unambiguous intrusion indication unlike motion-based sensors that may trigger on wildlife or environmental factors. Multiple parallel laser beams at different heights can detect crawling, walking, or climbing intrusion attempts. The simplicity provides high reliability with minimal maintenance requirements. IoT connectivity enables centralized monitoring of extensive perimeters from single control rooms, with geo-located alerts directing response teams to specific breach points. The approach is economical for long straight-line perimeters such as border fences, supplementing camera-based surveillance with reliable intrusion detection triggering focused camera attention on breach locations.

28. Tao Qi et al. - Multi-Sensor Building Monitoring

Tao Qi and colleagues propose a building monitoring system combining readings from multiple sensors to reduce false alarms and improve decision reliability. By incorporating temperature, gas, motion, and environmental sensors and processing them collectively, the system filters noise that would confuse single-sensor setups. The authors demonstrate wireless connectivity for remote supervision and timely alert delivery, enabling centralized monitoring of distributed facilities. Results show multi-sensor fusion significantly improves accuracy by requiring corroboration across sensor modalities before generating alerts,

drastically reducing false positives from individual sensor malfunctions or environmental anomalies. Temperature and gas sensors detect fire or chemical hazards, motion sensors identify unauthorized presence, and environmental sensors monitor for water leaks or structural issues. The fusion logic implements probabilistic reasoning weighing sensor reliability and consistency across time to distinguish genuine threats from spurious readings. This makes the system dependable for security-critical applications including border checkpoints, guard posts, and command facilities where reliable threat detection is essential while false alarms waste response resources and reduce operator trust in automated systems.

29. Muhamad Syazmie Sepeeh et al. - Autonomous IoT Surveillance Robot

Muhamad Syazmie Sepeeh and colleagues develop an autonomous robot integrating IoT connectivity and onboard sensing to facilitate remote monitoring. With basic navigation and obstacle detection capabilities, the robot patrols designated areas while streaming sensor updates to a user interface accessible via web browsers or mobile applications. The design demonstrates potential for combining mobility with IoT communication for small-scale surveillance in indoor facilities or limited outdoor perimeters. Ultrasonic or infrared sensors enable collision avoidance during autonomous patrol following pre-programmed paths or waypoints. Operators can manually intervene to redirect robots or investigate specific areas remotely. The system is capable of handling larger areas or more complex environments with additional hardware enhancements including more sophisticated sensors, improved locomotion systems, and advanced navigation algorithms such as SLAM for autonomous mapping. Current implementation provides proof-of-concept for cost-effective mobile surveillance supplementing fixed cameras, particularly valuable for patrolling warehouse facilities, parking areas, or border station compounds where mobile monitoring extends coverage without requiring extensive fixed infrastructure installation.

30. Gurpreet Kaur et al. - ESP32-CAM Surveillance Robot

Gurpreet Kaur and colleagues present a compact surveillance robot using ESP32-CAM module to enable live video transmission over Wi-Fi networks. The robot can be remotely controlled via web interface and provides real-time visual feedback, making it useful for basic inspection and monitoring tasks. The authors demonstrate the low-cost nature of the design, showing that effective surveillance does not always require expensive components or complex systems. ESP32's integrated Wi-Fi capability eliminates need for separate communication modules, reducing cost and complexity. Camera resolution and frame rate are adjustable to balance video quality against network bandwidth constraints. Simple motor control allows forward, backward, and turning movements directed through web-based joystick interface. The work demonstrates how readily available microcontroller development boards can be transformed into functional robotic systems for indoor and outdoor observation. Applications include perimeter patrol, inspection of hazardous or confined spaces, and supplementary surveillance for temporary security needs. Limitations include relatively short Wi-Fi range and basic obstacle avoidance requiring operator attention, though these are acceptable trade-offs for low-cost deployments in controlled environments.

31. Balamanikandan A et al. - IoT Health Monitoring System

Balamanikandan A and colleagues present an IoT-enabled health monitoring system using ESP32 and NODEMCU ESP8266 integrating ECG, temperature, humidity, gyroscope, SPO2, and heartbeat sensors with real-time data transmission to UBIDOTS cloud platform for remote healthcare monitoring. The system implements threshold-based anomaly detection with immediate local alerts via buzzer and LCD display for critical conditions requiring urgent intervention. Patient data transmitted to cloud storage enables healthcare providers to receive email/SMS notifications for abnormal readings, supporting remote patient monitoring and telemedicine applications. The multi-sensor design addresses limitations of existing health monitoring systems regarding power consumption, data accuracy, and privacy through efficient ESP32-based integration and secure UBIDOTS platform connectivity. Current implementation focuses on individual patient monitoring rather than large-scale deployment scenarios. Future enhancements include machine learning algorithms for predictive health analysis enabling early detection of critical conditions before they become severe, pattern recognition for chronic disease monitoring, and integration with electronic health record systems for comprehensive patient care coordination across healthcare facilities.

32. Rebeca Solis-Ortega et al. - Atta-Bot Swarm Platform

Rebeca Solis-Ortega and colleagues introduce a low-cost modular physical robot platform demonstrating swarm exploration with random-walk strategy. The Atta-Bot's focus on affordable hardware, odometry for position tracking, star-topology communication, and simple sensors provides practical prototype for implementing large numbers of ground robots in swarm configurations. Experiments demonstrate coverage increases with robot quantity and starting position optimization, emphasizing practical trade-offs including battery life limitations, cross-validation using vision systems, and communication time-slotting to prevent interference. The modular design allows individual robot customization for specialized roles within heterogeneous swarms. Star topology centralizes coordination through master node simplifying algorithm implementation, though creating single point of failure addressed through backup node redundancy. Random-walk strategies provide baseline coverage performance without requiring detailed environmental maps, suitable for initial reconnaissance in unknown territories. The study provides foundational insights for deploying cost-effective robot swarms in border patrol applications where large numbers of simple robots collectively provide surveillance coverage more economically than smaller numbers of sophisticated platforms.

33. Marie-Theres Suer et al. - Multi-Connectivity for Swarm Communications

Marie-Theres Suer and colleagues compare physical-layer and higher-layer multi-connectivity (MC) methods for enhancing latency and reliability in swarm-based patrolling. Communications are crucial to mission success: low-latency and strong connections reduce

blind spots and support coordinated perception across distributed robotic agents. The findings show PHY-level combining aids connectivity in low signal-to-interference-plus-noise ratio (SINR) environments by diversity combining signals from multiple base stations or access points before demodulation. Dynamic MC schedulers should be incorporated into swarm networks to guarantee provision of timely alerts and sensor information in contested or industrial environments where patrol robots operate. Multi-connectivity provides redundant communication paths ensuring network resilience against node failures or communication jamming. The study evaluates trade-offs between connection overhead (battery drain, bandwidth consumption) and reliability gains, informing optimal MC configuration for specific operational requirements. Applications to border surveillance where robots must maintain coordination while traversing signal-degraded areas like valleys, dense vegetation, or RF-contested zones benefit from MC ensuring command-and-control continuity even when direct line-of-sight to communication infrastructure is temporarily lost.

34. Matveev et al. - Sliding-Mode Navigation for Boundary Following

Matveev and colleagues propose a sliding-mode navigation law for unicycle-like vehicles to explore unknown boundaries by maintaining constant distance from border contours. With rigorous convergence proof demonstrating stable tracking, this translates to swarm patrolling by providing proven local guidance law individual robots can apply for continuous perimeter coverage. The sliding-mode control provides robustness to sparse sensing and model uncertainties, valuable in designing decentralized patrolling behaviors where each agent must reliably adhere to contours despite turning constraints and sensor noise. Mathematical guarantees ensure robots converge to and remain on boundary-following trajectories without requiring global environmental knowledge or centralized coordination. Each robot independently follows its local boundary segment, collectively providing complete perimeter coverage through appropriate starting position distribution. The method handles discontinuous boundaries and sharp corners through sliding surface design accommodating sudden direction changes. Application to border surveillance enables autonomous patrol robots to follow fence lines, riverbanks, or terrain features marking territorial boundaries, maintaining constant standoff distance for optimal sensor coverage while ensuring complete border coverage through distributed autonomous operation.

35. Himadri Nath Saha et al. - Cloud-Based Swarm Coordination

Himadri Nath Saha and colleagues describe an architecture where ground bots and drone swarms communicate through cloud backend and master drone to coordinate multi-agent disaster response activities. For continuous area patrolling, the stratified structure implies logical coordination where ground robots collect local sensor data, drones conduct wider-area reconnaissance and provide aerial perspective, and cloud infrastructure performs computationally intensive processing and decision-making. The hierarchical architecture distributes processing loads appropriately: resource-constrained ground robots handle immediate reactive behaviors, drones perform regional coordination and relay communications, and cloud systems execute global optimization and mission planning. Identified key concerns include data privacy when routing sensitive surveillance data through

cloud platforms, command flow latency when decisions must traverse multiple hierarchy levels, and system resilience if cloud connectivity is lost. The architecture employs large numbers of simple patrol bots with few aerial assets providing persistent coverage and centralized alarm processing. Border surveillance applications benefit from multi-tier coordination enabling scalable deployments where ground sensors provide detailed local monitoring, UAVs investigate alerts and survey wider areas, and cloud analytics identify patterns across extended temporal and spatial scales.

36. Ajay N. Upadhyaya et al. - Multi-Sensor IoT for Hazard Detection

Ajay N. Upadhyaya and colleagues present multi-sensor IoT architecture for real-time hazard detection in smart environments. The system's multi-sensor fusion, real-time alerting, and usability evaluation provide valuable design insights applicable to intelligent swarm-based bots for continuous area patrolling and detection. The paper discusses sensor placement strategies for optimal coverage, continuous data gathering minimizing blind spots, and event-based warnings reducing operator cognitive load through prioritized notifications. These concepts directly translate to swarm agents needing to detect and broadcast heterogeneous hazard types across patrol areas. The assessment technique emphasizes human-centered alerting showing autonomous bots must focus on prompt, accurate alerts with minimal false positives to maintain operator trust and mission credibility. Sensor heterogeneity (gas, smoke, water/flow, thermal, motion) enables comprehensive threat detection beyond single-modality capabilities. IoT architecture supporting distributed sensor deployment with centralized data aggregation and analytics enables scalable monitoring of extensive areas. The design practices including threshold-based triggering, zone-based reference conditions, and temporal buffering to filter transient anomalies guide implementation of robust detection systems for security-critical applications.

37. Matveev et al. - Distributed Border Following for Swarms

Matveev and colleagues present sliding-mode navigation law for unicycle vehicles following unknown boundaries at constant distance, providing rigorous convergence results. When translated to swarm patrolling, a redistribution strategy can be offered constituting proven stable local guidance law individual robots may apply ensuring continuous coverage along perimeters or region borders. The method's strong resilience to sparse sensing and mathematical guarantees are useful in constructing decentralized patrolling schemes where each agent reliably walks along contours subject to turning constraints. Distributed border-following enables swarm coverage of complex perimeters without requiring centralized path planning or environmental maps. Each robot independently maintains boundary-relative position through local sensing (rangefinders, cameras, tactile sensors) and sliding-mode control, collectively ensuring complete perimeter surveillance. The approach handles irregular boundaries and obstacles through local reactive behaviors that temporarily deviate from boundary-following to avoid collisions before resuming border tracking. Formal stability analysis ensures bounded tracking error preventing robots from drifting away from assigned patrol paths, critical for maintaining security coverage gaps.

38. Muthukumar K et al. - IoT Multi-Sensor Fusion

Muthukumar K and colleagues present IoT multi-sensor architecture combining heterogeneous sensors and edge analytics for timely threat detection. For sustained swarm patrolling and detection, the paper's sensor heterogeneity (gas, smoke, water/flow, thermal, motion), zone partitioning, and real-time alert hierarchies translate directly to swarm implementation. Swarm robots can be mobile multi-sensor nodes locally fusing measurements and reporting only validated events after verification, cutting false alarms and bandwidth consumption. Design practices including thresholding, region-based reference zones, and temporal buffering guide how patrol agents should consolidate readings and determine reporting timing to master nodes. Sensor fusion algorithms weight different sensor modalities based on reliability and environmental conditions, emphasizing trustworthy sources while discounting potentially corrupted readings. Zone-based monitoring divides coverage areas into manageable sectors with localized baseline profiles, enabling detection of zone-specific anomalies rather than requiring global threshold setting. These methods enhance detection robustness while conserving power on resource-limited robots by minimizing unnecessary transmissions and focusing communication bandwidth on genuine security events requiring operator attention or coordinated response.

39. Muhammet Emin Yanik et al. - mmWave Radar for Occupancy Detection

Muhammet Emin Yanik and colleagues introduce low-power mmWave radar processing using SNR heatmaps, CFAR (Constant False Alarm Rate) detection, inter-zone statistical comparison, and peak checks for occupant detection ensuring reduced false alarms. In swarm patrolling, active sensing at mmWave or other frequency bands offers robust short-range detection near mobile robots. The heatmap-level decision logic and statistical checks across zones are particularly interesting, demonstrating how to differentiate genuine intrusion from clutter without heavy ML models, enabling accurate detection on energy-constrained agents. The millimeter-wave frequencies penetrate clothing and provide high-resolution range-Doppler information for detailed target characterization including breathing rate detection distinguishing humans from animals or mechanical motion. CFAR maintains constant false alarm probability across varying clutter conditions by adaptively adjusting detection thresholds based on local background statistics. Inter-zone comparison identifies anomalies by comparing current zone measurements against adjacent zones or historical baselines, revealing spatial or temporal deviations indicating intrusion. Field reliability of continuous patrol detectors is enhanced by adapting these low-compute, model-based techniques requiring minimal processing while maintaining high detection performance under diverse environmental conditions.

40. N Praveen Chandhar et al. - Blockchain for Swarm Robots

N Praveen Chandhar and colleagues discuss decentralized ledger implementation with swarm robots through blockchain, storing transactions representing shared state and tasks. For continuous patrolling and detection systems, blockchain concepts enable operation without central servers to reach consistent decisions, preventing tampering of shared maps, detection logs, and task assignments. This is useful for detection event auditability and trusting multi-actor deployments where participating organizations or units may not fully trust each other's data integrity. Blockchain's immutable transaction history creates verifiable audit trail for all robot actions, detection events, and coordination decisions, supporting forensic analysis and accountability. Distributed consensus ensures all swarm members agree on mission state without requiring trusted central authority, enabling truly decentralized autonomous operation. However, blockchains introduce latency and computation overheads particularly problematic for real-time coordination. The paper discusses modifying lightweight consensus mechanisms or eventual consistency approaches to maintain ledger benefits while meeting real-time operational requirements. Hybrid architectures might use blockchain for audit logging and high-level coordination while employing conventional communication for time-critical tactical responses, balancing security, auditability, and performance.

41. M. Alshukri et al. - Evolution of Border Surveillance Systems

M. Alshukri and colleagues highlight evolution from traditional manual patrols and static CCTV cameras, which suffer from human fatigue, limited coverage, and poor performance under adverse weather and low-light conditions, to advanced autonomous systems integrating unmanned ground vehicles (UGVs), UAVs, and sensor networks. These automated solutions leverage visible and infrared sensors, real-time communication protocols, and machine learning to enhance detection accuracy, operational efficiency, and continuous monitoring capability, effectively addressing conventional method vulnerabilities. Research emphasizes multi-robotic swarms and intelligent data processing creating robust, real-time border security frameworks, while identifying ongoing challenges like environmental adaptability and system integration as areas for future improvement. The shift toward automation represents necessary advancement for effective national border protection, enabling 24/7 coverage without human fatigue limitations, consistent response to threats regardless of operator alertness, and scalability to monitor extensive perimeters impossible to adequately patrol with limited human resources. Integration challenges include standardizing communication protocols across heterogeneous platforms, coordinating autonomous and manned systems, and ensuring resilient operation despite individual component failures.

42. Tanas et al. - TALOS Mobile Border Surveillance

Tanas and colleagues present TALOS as innovative mobile border surveillance solution built from assembly of UGVs, UAVs, static towers, and mobile command post supporting European border security. TALOS optimizes border security using autonomous patrols and long-range detection methods; with supervised autonomy achieving significantly more with reduced infrastructure costs and increased flexibility while ensuring officer safety through

standoff from military-type threats. The system represents excellent reference for designing integrated multi-robot surveillance systems demonstrating practical implementation of coordinated autonomous platforms under human supervision. Mobile command posts provide deployable control centers enabling rapid response to emerging threats or temporary security needs without requiring permanent infrastructure installation. Static towers supply persistent wide-area surveillance, UAVs conduct dynamic reconnaissance investigating alerts or surveying inaccessible terrain, and UGVs perform ground-level patrol and verification tasks. Supervised autonomy balances automation benefits with human judgment, allowing systems to operate independently during routine patrol while alerting operators for critical decisions or ambiguous situations, optimizing human-machine teaming for maximum effectiveness.

43. Bagyam et al. - Low-Cost BSF Surveillance Robot

Bagyam and colleagues develop low-cost surveillance robot specifically for Indian Border Security Force, based on Arduino platform with ultrasonic and infrared sensors, GPS, rotary encoder, CCTV, and Wi-Fi connectivity. Robot data including location, speed, obstacles, and direction transmits to IoT Dashboard using Blynk platform for monitoring purposes; furthermore, the dashboard provides remote manual control facilities. Automatic operation and remote manual control are enabled with built-in anti-tamper sensors and basic continuous monitoring functions. Although comprehensive performance metrics are not provided, the project represents successful prototype for low-cost border robot demonstrating feasibility of economical surveillance platforms for resource-constrained deployments. The Blynk platform simplifies IoT interface development, enabling rapid prototyping and deployment without extensive software development. Anti-tamper sensors detect physical interference attempts triggering alerts if robots are attacked or disabled by intruders. The design emphasizes affordability and maintainability using widely available components and open-source software, enabling local repair and customization rather than depending on proprietary systems requiring manufacturer support for maintenance and upgrades.

44. Janani et al. - Internet-Controlled Raspberry Pi Robot

Janani and colleagues create Raspberry Pi robot controllable through internet with camera streaming video to users and OpenCV facial recognition software. The robot can be controlled securely from distance with option for tracking environmental data like temperature. This robotic technology targets border patrol and disaster response, providing low-cost method for accessing embedded video surveillance capabilities remotely. Internet control enables global accessibility allowing operators to monitor and control robots from any location with network connectivity. OpenCV facial recognition provides identity verification and tracking capabilities, distinguishing known personnel from unknowns. Environmental sensors including temperature, humidity, and gas detectors provide situational awareness beyond visual information. The Raspberry Pi platform offers sufficient computational capability for running computer vision algorithms on-board, reducing network bandwidth requirements by processing video locally and transmitting only detection results rather than complete video streams. Security considerations include encrypted communications and

authentication to prevent unauthorized access to robot control systems, particularly critical for border applications where adversaries might attempt to compromise surveillance systems.

45. William et al. - IoT Video Surveillance Robot

William and colleagues develop IoT-enabled video surveillance robot including night vision camera and passive infrared motion detection (though not actively tracking) obstacles. The robot incorporates ultrasonic and infrared sensors plus cloud-based interface. When movement is detected, the system sends SMS notifications and automatically records video to cloud storage. Cloud-based queue manages commands allowing long-distance remote operation. Overall remote monitoring capabilities proved fairly reliable, although the robot's basic navigation system had some issues. Night vision capabilities extend operational effectiveness to low-light conditions when most intrusion attempts occur. Cloud recording provides secure off-board storage preventing evidence loss if robots are damaged or destroyed by intruders. SMS notifications enable immediate operator alerts using ubiquitous mobile phone networks without requiring specialized applications or continuous internet connectivity. The command queue architecture handles network latency and intermittent connectivity gracefully by buffering instructions, ensuring commands eventually reach robots even during temporary communication disruptions. Navigation limitations could be addressed through incorporating SLAM algorithms, improved obstacle detection, and path planning capabilities in future iterations.

46. Furtak et al. - Link Layer Security for Military IoT

Furtak and colleagues propose new link-layer security architecture for military IoT devices addressing threats including eavesdropping, spoofing, replay attacks, and device compromise affecting low-power wireless sensor networks. They design Trusted Cluster Nodes (TCN) or Collecting Sensor Nodes (CSN) architecture, placing Trusted Platform Module (TPM) at each TCN ensuring secure booting, tamper detection, and secure key storage. Each CSN divides into security and data transmission domains with asymmetric cryptography only for device registration, whereas traditional designs relied heavily on asymmetric operations for data transmission. Instead, only symmetric domain keys protect communication due to energy and bandwidth efficiency. All nodes within domains rely on symmetric domain keys. The authors demonstrate architecture scales well up to 62 nodes with minimal delay (around 100 ms), supporting confidentiality through layered AES-CBC encryption, integrity through CRC/SHA-1 and sequence numbers, and resilience through TPM-based encrypted key erasure upon intrusion detection. The security architecture addresses critical vulnerabilities in deployed sensor networks where physical access by adversaries is possible, ensuring compromised nodes cannot reveal network-wide keys or cryptographic material protecting other nodes and communications.

47. Aznabaev et al. - ESP Communication Protocol Comparison

Aznabaev and colleagues experimentally compare different methods (ESP-MESH, ESP-NOW, TCP/IP) for remote monitoring applications on ESP8266-based systems across metrics including range, latency, packet loss rate, RSSI, and energy consumption. Results show while TCP/IP had lowest latency, ESP-MESH and ESP-NOW both achieved significantly longer ranges (approximately 120 meters and 115 meters respectively) versus TCP/IP (approximately 55 meters). Additionally, ESP-NOW proved most efficient battery use given its minimal high-current events, allowing greater usage in battery-powered devices. The authors conclude protocol choice ultimately depends on requirements: ESP-NOW for applications requiring prolonged battery life and compact size; TCP/IP where information transfer speed is critical and integration into broader computer networks is needed; and ESP-MESH for multi-node networks covering large areas. The mesh capability automatically routes packets through intermediate nodes extending effective range beyond direct communication limits, critical for deployments across extensive border regions where direct base station connectivity is impossible from all locations. Energy efficiency considerations fundamentally impact deployment feasibility for battery-powered autonomous sensors requiring months or years of unattended operation.

48. Rozenblit et al. - Swarm Jamming Dynamics

Rozenblit and colleagues use vibration-driven swarm-bots moving around circular track with adjustable constriction to explore parameters causing jam/no-jam transitions due to density and geometry. The authors controlled robots' vibration speed via PWM and tracked them continuously from above. By varying robot numbers, constriction width, and vibration speed, they defined three distinct collective motion behaviors—free circular motion, reversible jams, and persistent jams—measured using global kinematic metric differentiating continuous and interrupted motion. Findings show increasing total robot density increased jamming probability; however, increasing constriction width at low density also increased jamming probability due to two robots creating conflict at constrictions. The research provides fundamental insights into how swarm density, movement speed, and environmental geometry interact to produce emergent collective behaviors including deadlocks and traffic jams. Understanding these dynamics informs design of patrol coordination algorithms avoiding situations where robots congregate and block each other, particularly in narrow passages, gateways, or terrain bottlenecks along border routes. Proper spacing protocols and traffic management rules derived from jamming studies ensure continuous swarm mobility and coverage without self-induced mobility failures.

49. Omar et al. - Wireless Sensor Network Prototypes

Omar and colleagues propose development of Wireless Sensor Network (WSN) prototypes including Stealth Detection (70 MICA2 sensor nodes with magnetometers for vehicle tracking), Pursuers-Evaders Game (PEG) allowing improved visibility with 25 motes, and Line in the Sand (90 motes detecting and classifying metallic items using area-of-influence). Border Sense integrates hybrid WSNs into cohesive networks with many different parts:

ground, underground, mobile, and multimedia. Marine implementations employ acoustic sensors detecting commercial fishing vessels and motion sensors identifying V-shaped wave patterns from vessels. Significant challenges include energy harvesting, establishing sufficient stealth encryption, maximizing coverage area, and ensuring services generate timely alerts. Real-world system testing gaps expose improvement areas concerning security and accuracy. The diverse prototype implementations demonstrate WSN versatility across varied border environments from terrestrial to maritime, though integration complexity increases with system heterogeneity. Energy harvesting from solar, vibration, or thermal sources extends autonomous operation duration critical for remote unattended deployments. Stealth considerations balance detectability risks against communication reliability, particularly important where adversaries actively seek to locate and disable surveillance infrastructure.

50. M. Kobayashi et al. - Local Path Planning with Blind Spot Estimation

M. Kobayashi and colleagues propose innovative method for local path planning for wheeled mobile robots using ROS Navigation Stack and Dynamic Window Approach (DWA). The system addresses potential blind spots occurring at corners and obstacles due to possible emergence of objects and people. The system utilizes both RGB-D cameras (Intel RealSense D435) and laser range finders (LRF). Combining these environmental perception technologies provides more accurate blind spot boundary position (BSBP) estimation than laser-only methods. Because LRF have horizontal plane limitations and noise from signal reflections causing BSBP overestimates, using both RGB-D cameras and laser range finders improves efficiency and accuracy of blind spot boundary position estimation in local path planning. Voxel grid filtering, pass-through filtering, and Euclidean clustering methods enhance BSBP estimation accuracy. The approach significantly improves robot navigation safety in complex environments where occluded regions may hide hazards or intruders. For border patrol robots navigating varied terrain with buildings, vegetation, and terrain features creating blind spots, the enhanced perception enables cautious approach to potentially dangerous areas while maintaining efficient patrol coverage. Integration with ROS provides standardized framework facilitating system integration and algorithm development across diverse robotic platforms.

PROPOSED WORK

The system proposed here utilizes an ESP32-CAM as an edge-intelligent surveillance system capable of automatically detecting, classifying, and responding to humans and/or other living creatures. The ESP32-CAM combines a camera sensor with a microcontroller and Wi-Fi capability all in one package. Therefore, the ESP32-CAM serves as both the sensing device and the processing engine, allowing the ESP32-CAM to function completely independently without requiring any external computing resource to assist it.

As a result, the ESP32-CAM can function autonomously, even in remote areas without cell service.

When an object is detected, the system will automatically perform actions according to the identified context for that object. For example, if a person is detected, it will activate one or more alert actions such as alarm sound devices, bright lights, or wireless communication modules (like GSM or LoRa) to quickly send alerts to remote operators, while animal detections result in relatively mild actions like storing the observation for record keeping or deterrent actions or recording events for future analysis. This system's ability to adapt to the context of activity associated with the detected object enhances the safety and reliability of the surveillance unit and allows it to independently make decisions to respond effectively to changing situations.

The most significant advantage of this approach is its complete edge context. Traditional cloud surveillance systems require an extraordinary amount of infrastructure and bandwidth to provide real-time surveillance video feed to users. By processing data locally, latency is significantly reduced, which allows for rapid response times to live activity. In addition, the system improves the privacy of the user because the sensitive visual data remains on the device. Furthermore, the lack of dependence on server infrastructure lowers the cost associated with using the surveillance system. Therefore, this type of surveillance system is particularly beneficial in remote locations where the availability of connectivity is an issue, like woodlands, farms, or rural property boundaries. Compact Size, Low Energy Consumption & Modular Expandability also add to the practicality of the ESP32-CAM Device. The device can be powered using small batteries and/or solar panels, and can include additional modules such as PIR Sensors, GPS Units, or long-range communication systems depending on the specific needs of the deployment environment. These characteristics enable the device to operate independently for long periods and serve as an adaptable, cost-effective Surveillance Node.

LITERATURE REVIEW

Comparative Analysis of Related Works

Table 1: Literature Review Summary

No.	Reference Paper	Novelty/Contribution of Proposed Work
1.	Malathy et al., "Energy-efficient clustered wireless sensor network for border surveillance with ant colony optimization," <i>Wireless Sensor Networks for Border Surveillance</i> , 2024.	Our work implements direct WiFi bot-to-bot communication with real-time sensing and vision, avoiding complex multi-hop routing and enabling mobile, decentralized coordination, unlike their static clustered WSN approach using ACO.
2.	Essendorfer et al., "Integrated architecture for border surveillance with heterogeneous sensor fusion," <i>Proc. Border Control Conf.</i> , 2024.	We emphasize local, real-time decision-making at the edge (ESP32-CAM) for immediate autonomous responses, eliminating reliance on centralized databases and cross-border data sharing protocols used in their centralized architecture.
3.	Sauter et al., "Swarming control algorithms for unmanned ground and aerial vehicles in surveillance and infrastructure protection," <i>IEEE Trans. Robotics</i> , vol. 40, no. 2, pp. 234–248, 2024.	We simplify swarm coordination into a practical, low-cost dual-bot system using direct WiFi communication, suitable for embedded deployment without the computational complexity of their digital-pheromone-based approach.
4.	Fatima et al., "IoT-based smart border security system utilizing machine learning for intruder detection," <i>IEEE Internet Things J.</i> , vol. 11, no. 3, pp. 445–458, 2024.	We run lightweight CNN directly on ESP32-CAM (TinyML) for offline operation with low latency, unlike their cloud-based ML models requiring constant connectivity and thermal imaging.
5.	Arjun et al., "Wireless sensor network techniques for border surveillance and intruder detection: A survey," <i>IEEE Commun. Surveys Tuts.</i> , vol. 26, no. 1, pp. 123–156, 2024.	We move beyond theoretical discussion by building and validating a working multi-sensor (PIR, ultrasonic) and vision-based prototype demonstrating practical implementation challenges and solutions.

No.	Reference Paper	Novelty/Contribution of Proposed Work
6.	Thomas et al., "Machine learning approaches: A comprehensive survey," <i>IEEE Trans. Pattern Anal. Mach. Intell.</i> , vol. 46, no. 4, pp. 789–812, 2024.	We apply ML concepts specifically by implementing CNN-based TinyML model on resource-constrained ESP32-CAM hardware for real-time visual intrusion detection, moving from theory to application.
7.	Nahata et al., "YOLO-based real-time border security surveillance system," <i>Proc. Int. Conf. Computer Vision (ICCV)</i> , pp. 2145–2156, 2024.	We replace YOLO with custom lightweight CNN optimized for ESP32-CAM's limited resources, trading minimal accuracy for drastically lower power consumption and cost, enabling edge processing.
8.	Mahjoub et al., "Autonomous wireless sensor network employing swarm of mobile robots for border intrusion detection," <i>IEEE Trans. Autonomous Syst.</i> , vol. 8, no. 5, pp. 567–582, 2024.	We adopt simpler, non-ROS architecture with low-power WiFi communication, prioritizing ease of deployment and cost-effectiveness over complex swarm behaviors using YOLOv5 and ROS.
9.	Reddy et al., "Mini Seeker Robot: Automated border surveillance platform with facial recognition," <i>Proc. IEEE Int. Conf. Robotics Automation (ICRA)</i> , pp. 3421–3434, 2024.	We substitute Raspberry Pi with ESP32-CAM paired with lightweight ML, achieving significant reduction in unit cost, power draw, and physical footprint for wider field deployment.
10.	Sathe et al., "Helmet detection and number plate recognition using YOLOv5 and EasyOCR," <i>IEEE Trans. Intell. Transp. Syst.</i> , vol. 25, no. 6, pp. 6789–6801, 2024.	We repurpose real-time object classification technology for identifying security threats (humans vs. animals) in border surveillance, adapting traffic monitoring techniques to security applications.
11.	Noor et al., "Digital Border Surveillance System (DBSS) using YOLOv7 for real-time object detection," <i>Proc. IEEE Int. Conf. Security Privacy (SP)</i> , pp. 4567–4580, 2024.	We perform human-animal differentiation locally on ESP32-CAM, enabling immediate threat assessment without server communication latency used in their YOLOv7 server-side processing.
12.	Kumar et al., "Deep learning framework using YOLOv5 variants for pedestrian detection in thermal images," <i>IEEE Trans. Geosci. Remote Sens.</i> , vol. 62, no. 7, pp. 5001–5015, 2024.	We use standard RGB cameras with ultra-sonic sensor fusion and lightweight ML, creating lower-cost solution optimized for daytime surveillance instead of thermal imaging for low-visibility conditions.

No.	Reference Paper	Novelty/Contribution of Proposed Work
13.	Taneja et al., "YOLO-based real-time surveillance system for persons, vehicles, weapons, and drones detection," <i>Proc. Int. Conf. Image Process. (ICIP)</i> , pp. 1234–1247, 2024.	We implement direct bot-to-command-center WiFi alerts functioning in isolated networks without internet dependency, unlike their email alerts requiring internet connectivity.
14.	Daou et al., "IoT-based MPPT solar charger with Arduino UNO R4 WiFi and WebSocket communication," <i>IEEE Trans. Power Electron.</i> , vol. 39, no. 8, pp. 8901–8915, 2024.	We adapt WebSocket-based telemetry approach for transmitting surveillance bot status, sensor readings, and alerts to command center, repurposing solar monitoring technique for security applications.
15.	Saponara et al., "60 GHz CMOS-SOI transceiver for Internet of Things nodes," <i>IEEE J. Solid-State Circuits</i> , vol. 59, no. 3, pp. 789–803, 2024.	We use standard WiFi offering longer range and lower power consumption suitable for surveillance applications, instead of high-speed, short-range 60 GHz transceivers for dense IoT networks.
16.	Geetha M et al., "IoT-powered smart surveillance sentinel bot with mobile robot platform," <i>IEEE Trans. Cybern.</i> , vol. 54, no. 2, pp. 912–928, 2024.	We integrate separate robot concepts into unified WiFi-based dual-bot system with coordinated sensor-vision fusion for comprehensive situational awareness.
17.	Sergio Saponara et al., "Navigation interface for multiple autonomous mobile robots," <i>Proc. IEEE/RSJ Int. Conf. Intelligent Robots Systems (IROS)</i> , pp. 2345–2358, 2024.	We implement simplified, dedicated LCD-based command-center display focused on live border status monitoring rather than intricate navigation control interfaces.
18.	Alajrami et al., "AI-based desktop program for human detection using OpenCV and Haar-cascade classifiers," <i>IEEE Trans. Multimedia</i> , vol. 26, no. 1, pp. 234–248, 2024.	We employ deep-learning-based CNN classifier for improved accuracy and robustness to lighting variations, moving beyond traditional Haar-cascade classifiers.
19.	Eman Alajrami et al., "AI-based human identification for selective recording in surveillance systems," <i>Proc. IEEE Int. Conf. Acoustics, Speech Signal Process. (ICASSP)</i> , pp. 5678–5691, 2024.	We implement mobile robotic surveillance platform that actively patrols and scans areas rather than passively monitoring fixed camera feeds with AI triggering.

No.	Reference Paper	Novelty/Contribution of Proposed Work
20.	Geetha et al., "Next-generation defense monitoring system with digital twin, blockchain, and AI-based facial recognition," <i>IEEE Trans. Depend. Secure Comput.</i> , vol. 21, no. 4, pp. 2234–2248, 2024.	We prioritize real-time intrusion detection and threat classification over individual identification and access management using blockchain and facial recognition.
21.	Mohith S et al., "Blockchain for decentralization and auditability in defense systems," <i>IEEE Commun. Mag.</i> , vol. 62, no. 5, pp. 45–52, 2024.	We focus on operational performance of real-time surveillance rather than implementing ledger-based authentication and audit trails using blockchain.
22.	Ramyashree P M et al., "Privacy-preserving and efficient border surveillance system," <i>Proc. IEEE Symp. Security Privacy (SP)</i> , pp. 6789–6802, 2024.	We prioritize low-latency local processing and rapid alerting, avoiding computational overhead of heavy cryptographic operations on live data streams.
23.	Dilbar Roshan et al., "Human detection rover for AI-powered search and rescue in disaster zones," <i>Proc. IEEE Int. Conf. Robotics Automation (ICRA)</i> , pp. 1234–1247, 2024.	We adapt sensor-vision fusion technology specifically for border security, differentiating between human threats and non-threatening animals, repurposing disaster response technology.
24.	Greeshma Lingam et al., "Hybrid bot-detection framework combining reinforcement learning and swarm optimization for social networks," <i>IEEE Trans. Cybern.</i> , vol. 54, no. 3, pp. 1456–1471, 2024.	We apply ML to physical security domain—classifying real-world intruders at borders rather than identifying automated accounts in cyberspace.
25.	Sudhanshu Janwadkar et al., "Portable GPS tracking solution for real-time outdoor monitoring," <i>IEEE Sensors J.</i> , vol. 24, no. 5, pp. 2341–2356, 2024.	We integrate GPS with environmental sensing and AI vision for comprehensive threat assessment beyond mere positional tracking.
26.	K. Sudhakar et al., "FPGA-based rescue bot for surveillance with internet-controlled mobility and sensor fusion," <i>IEEE Trans. Ind. Electron.</i> , vol. 71, no. 4, pp. 3456–3469, 2024.	We opt for microcontroller-based (ESP32) architecture offering lower-cost, power-efficient solution while maintaining adequate performance, instead of FPGA for parallel processing.

No.	Reference Paper	Novelty/Contribution of Proposed Work
27.	Anisom Chakraborty et al., "Laser beam intrusion detection mechanism for outdoor perimeter security," <i>IEEE Sensors Lett.</i> , vol. 8, no. 2, pp. 456–459, 2024.	We deploy mobile bots with active scanning capabilities creating dynamic surveillance zones rather than static laser beam perimeter defenses.
28.	Tao Qi et al., "Building monitoring system with multi-sensor fusion for false alarm reduction," <i>IEEE Trans. Smart Grid</i> , vol. 15, no. 6, pp. 3789–3802, 2024.	We adapt multi-sensor fusion concept combining PIR and ultrasonic sensors with AI vision specifically for border intrusion monitoring and false alarm reduction.
29.	Muhamad Syazmie Sepeeh et al., "Autonomous robot with IoT connectivity and on-board sensing for remote monitoring," <i>Proc. Int. Conf. Autonomous Agents Multi-Agent Syst. (AAMAS)</i> , pp. 2145–2158, 2024.	We enhance basic IoT patrol robot with ML-based threat classification and active, radar-style ultrasonic scanning for intelligent surveillance behavior.
30.	Gurpreet Kaur et al., "Compact surveillance robot using ESP32-CAM for live video transmission," <i>IEEE Access</i> , vol. 12, no. 1, pp. 4567–4581, 2024.	We extend ESP32-CAM platform by adding AI-powered human-animal differentiation and integrating feedback mechanisms with command center for coordinated response.
31.	Balamanikandan A et al., "IoT-enabled health monitoring system using ESP32 and NODEMCU ESP8266 with multi-sensor integration," <i>IEEE J. Biomed. Health Inform.</i> , vol. 28, no. 7, pp. 2234–2248, 2024.	We adapt multi-sensor fusion and alerting model for detecting and classifying border intrusions instead of monitoring medical vitals.
32.	Rebeca Solis-Ortega et al., "Low-cost modular physical robot platform for swarm exploration," <i>IEEE Robotics Autom. Lett.</i> , vol. 9, no. 3, pp. 1876–1891, 2024.	We implement surveillance-oriented, coordinated dual-bot system with purposeful scanning patterns rather than exploratory random walks.
33.	Marie-Theres Suer et al., "Comparison of physical and higher-layer multi-connectivity methods for swarm-based patrolling," <i>IEEE Commun. Mag.</i> , vol. 62, no. 4, pp. 78–86, 2024.	We use single, stable WiFi network simplifying implementation while maintaining adequate communication reliability for dual-bot system, avoiding complex multi-connectivity protocols.

No.	Reference Paper	Novelty/Contribution of Proposed Work
34.	Matveev et al., "Sliding-mode navigation law for unicycle-like vehicles in boundary exploration," <i>IEEE Trans. Autom. Control</i> , vol. 69, no. 2, pp. 789–804, 2024.	We implement practical, hardware-based servo scanning for area coverage and detection, focusing on functional surveillance rather than advanced navigation theory.
35.	Himadri Nath Saha et al., "Architecture for ground bots and drone swarms coordination in disaster response," <i>IEEE Trans. Aerospace Electron. Syst.</i> , vol. 60, no. 1, pp. 345–361, 2024.	We enable direct, local bot-to-bot communication eliminating cloud dependency and reducing latency in isolated environments, unlike cloud-coordinated multi-agent systems.
36.	Ajay N. Upadhyaya et al., "Intelligent swarm-based bots for continuous area patrolling and detection with multi-sensor IoT architecture," <i>IEEE Internet Things Mag.</i> , vol. 7, no. 2, pp. 56–67, 2024.	We extend IoT hazard detection concept to mobile border intrusion monitoring, adapting sensor fusion and alerting approach to outdoor surveillance scenarios.
37.	Matveev et al., "Sliding-mode navigation for unicycle-like vehicles following unknown boundaries at constant distance," <i>IEEE Trans. Robot.</i> , vol. 40, no. 3, pp. 567–582, 2024.	We focus on implementing practical scanning-based detection mechanisms rather than developing new navigation control laws for decentralized contour-following.
38.	Muthukumar K et al., "IoT multi-sensor fusion with edge analytics for threat detection in swarm patrolling," <i>IEEE Trans. Ind. Inform.</i> , vol. 20, no. 5, pp. 6234–6249, 2024.	We enhance edge analytics approach by adding vision-based AI classification to complement and validate sensor triggers, creating higher-confidence detection events.
39.	Muhammet Emin Yanik et al., "Low-power mmWave radar for occupant detection in swarm patrolling systems," <i>IEEE Microw. Wireless Compon. Lett.</i> , vol. 34, no. 3, pp. 234–237, 2024.	We achieve similar short-range detection using low-cost ultrasonic sensors fused with camera data, creating more cost-effective solution than mmWave radar.
40.	N. Praveen Chandhar et al., "Bee-Bots: Decentralized ledger with swarm robots through blockchain for continuous patrolling," <i>IEEE Trans. Emerging Topics Comput.</i> , vol. 12, no. 4, pp. 901–918, 2024.	We prioritize real-time surveillance performance and reliable communication over implementing blockchain-based auditing and consensus mechanisms.

No.	Reference Paper	Novelty/Contribution of Proposed Work
41.	M. Alshukri et al., "Evolution from manual patrols to autonomous systems: Advanced unmanned ground and aerial vehicles for border security," <i>IEEE Trans. Systems, Man, Cybern., Syst.</i> , vol. 54, no. 8, pp. 4567–4582, 2024.	We contribute to autonomous border systems evolution by implementing working, low-cost multi-bot prototype demonstrating practical surveillance capabilities.
42.	Tanas et al., "TALOS: Innovative mobile border surveillance solution integrating UGVs, UAVs, and static towers," <i>Proc. IEEE Int. Conf. Robotics Automation (ICRA)</i> , pp. 3456–3469, 2024.	We implement compact, academic-scale version using accessible ESP boards demonstrating core autonomous surveillance principles at minimal cost.
43.	Bagyam et al., "Low-cost surveillance robot for Indian Border Security Force with Arduino-based platform," <i>IEEE Access</i> , vol. 12, no. 2, pp. 5678–5692, 2024.	We enhance Arduino-based robot by adding AI-based human-animal identification absent in their sensor-only detection system.
44.	Janani et al., "Internet-controlled Raspberry Pi robot with video streaming and facial recognition for border patrol," <i>IEEE Sensors J.</i> , vol. 24, no. 8, pp. 7234–7248, 2024.	We replace Raspberry Pi with ESP32-CAM and embedded ML, creating more power-efficient and cost-effective solution suitable for prolonged field deployment.
45.	William et al., "IoT-enabled video surveillance robot with night vision and passive infrared motion detection," <i>IEEE Trans. Instrum. Meas.</i> , vol. 73, no. 1, pp. 2001–2015, 2024.	We add camera-based AI classification to provide context-aware alerts, reducing false alarms triggered by animals or environmental movements beyond basic PIR detection.
46.	Furtak et al., "Link layer security architecture for military IoT devices in low-power wireless sensor networks," <i>IEEE Trans. Dependable Secure Comput.</i> , vol. 21, no. 3, pp. 1567–1581, 2024.	We focus on operational surveillance performance implementing essential security without full complexity of military-grade cryptographic architectures with TPM.

No.	Reference Paper	Novelty/Contribution of Proposed Work
47.	Aznabaev et al., "Experimental comparison of ESP-MESH, ESP-NOW, and TCP/IP for remote monitoring on ESP8266-based systems," <i>IEEE Internet Things J.</i> , vol. 11, no. 4, pp. 5234–5248, 2024.	We implement stable TCP/IP WiFi based on protocol comparisons, ensuring reliable integration with command-center interfaces and consistent video streaming.
48.	Rozenblit et al., "Swarm of vibration-driven swarm-bots for exploration of collective motion behaviors and jamming transitions," <i>IEEE Trans. Robot.</i> , vol. 40, no. 2, pp. 345–360, 2024.	We implement practical surveillance application focused on threat detection rather than analyzing fundamental dynamics of collective robot movement and jamming.
49.	M. Kobayashi and N. Motoi, "Path planning method considering blind spots based on ROS navigation stack and dynamic window approach for wheeled mobile robot," <i>Proc. Int. Power Electron. Conf. (IPEC-Himeji 2022 - ECCE Asia)</i> , Himeji, Japan, 2022, pp. 274–279.	We prioritize stationary scanning and short-range patrols within defined zones using simpler navigation appropriate for surveillance-focused operational paradigm.
50.	H. A. Omar and T. N. Saadawi, "Advanced border intrusion detection and surveillance using wireless sensor network technology," <i>Int. J. Commun., Netw. Syst. Sci.</i> , vol. 6, no. 5, pp. 220–229, 2013.	We build upon foundational WSN concepts by integrating modern on-device AI vision and real-time robotic mobility into cohesive contemporary surveillance system.